

25th European Conference on Computational Biology (ECCB)

Tutorial session T13: 13:45 -17:30

**Democratizing multi-omics data analysis
with BiomiX and Nextflow: from single-
omics to advanced integration**

Presenters :

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Istituto Nazionale dei Tumori, Milan,
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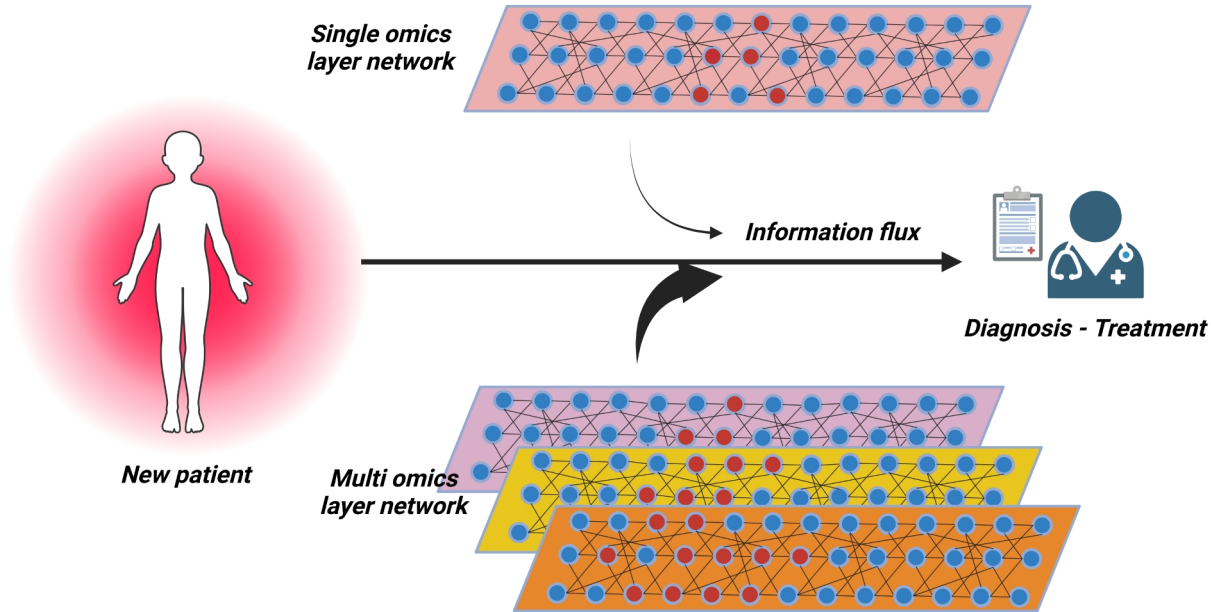


BIOMIX

MULTIOMICS DATA INTEGRATION



Multi-omics data integration, the grail of biology understanding ?



Pros

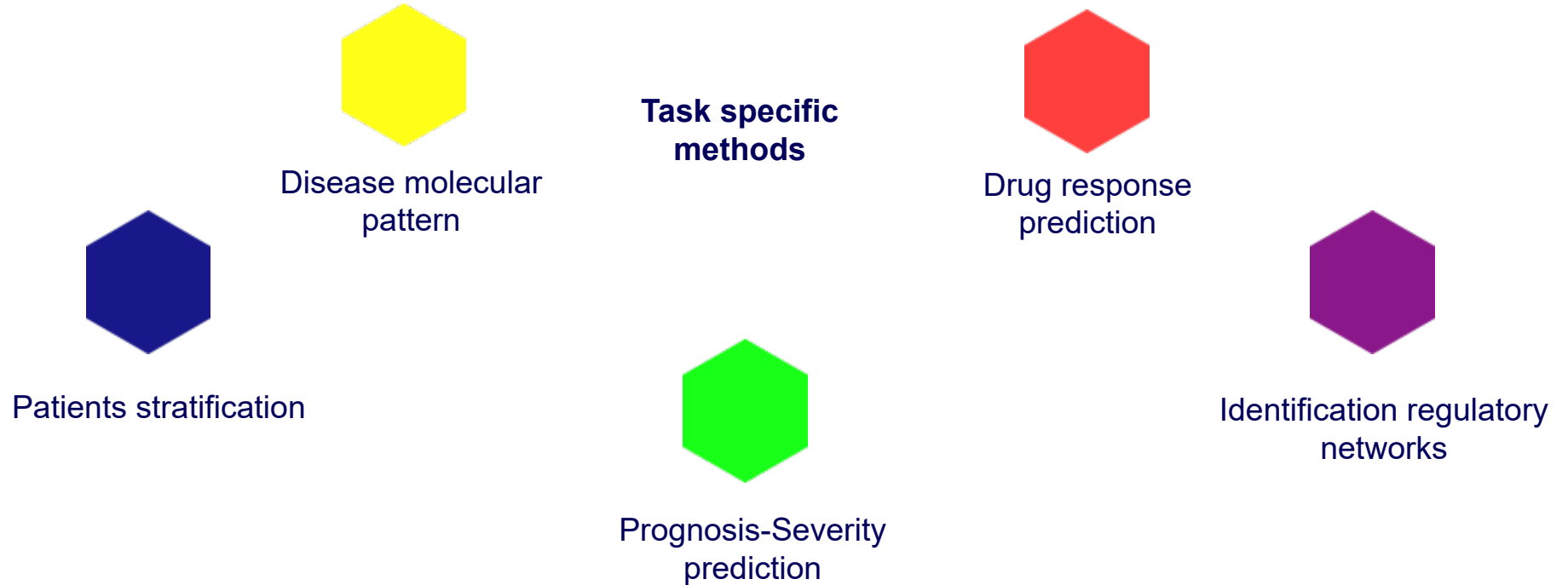
- Observations strengthened
- Holistic interpretation
- Detailed characterization

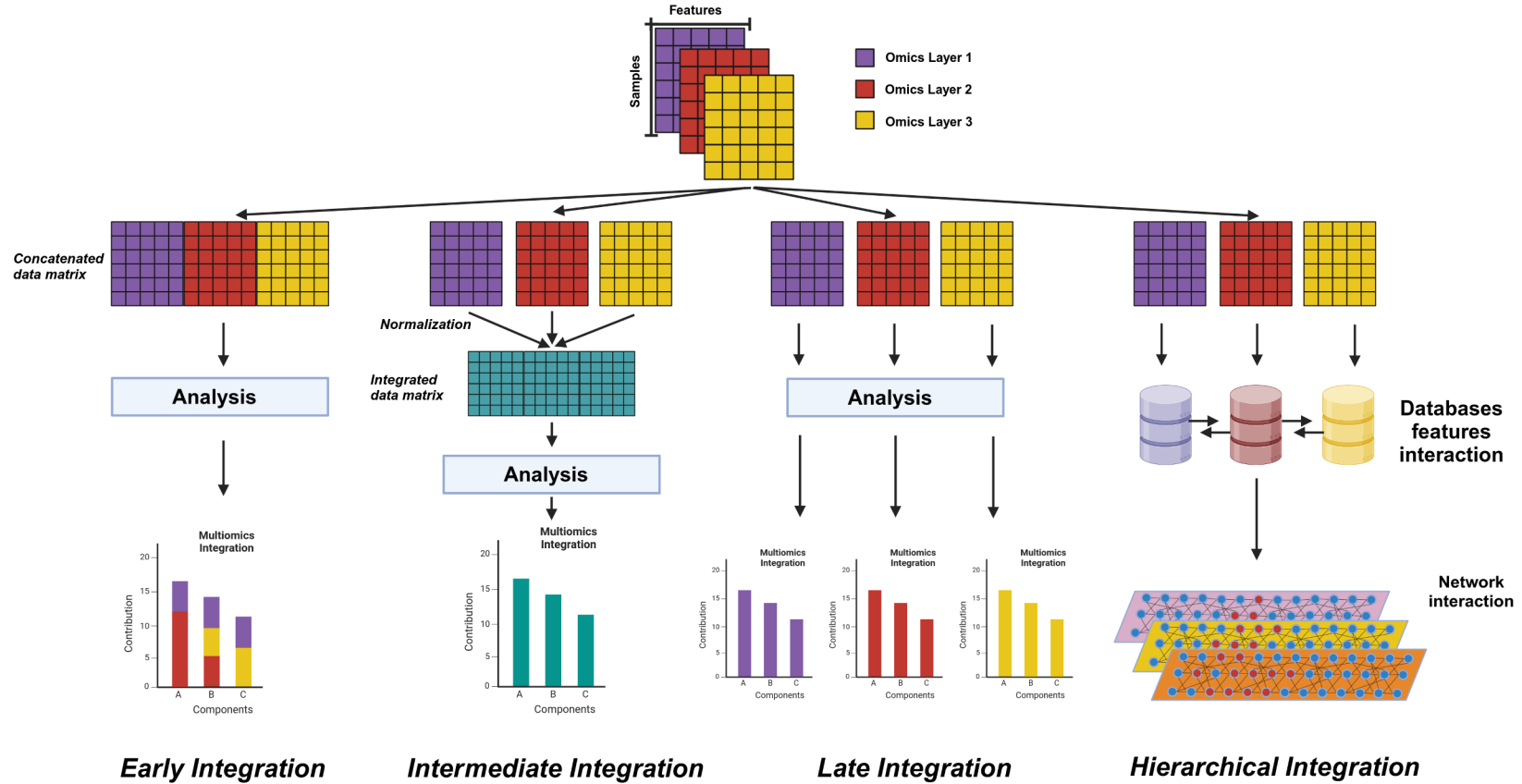
Cons

- Different distribution
- Course of dimensionality
- Missing data



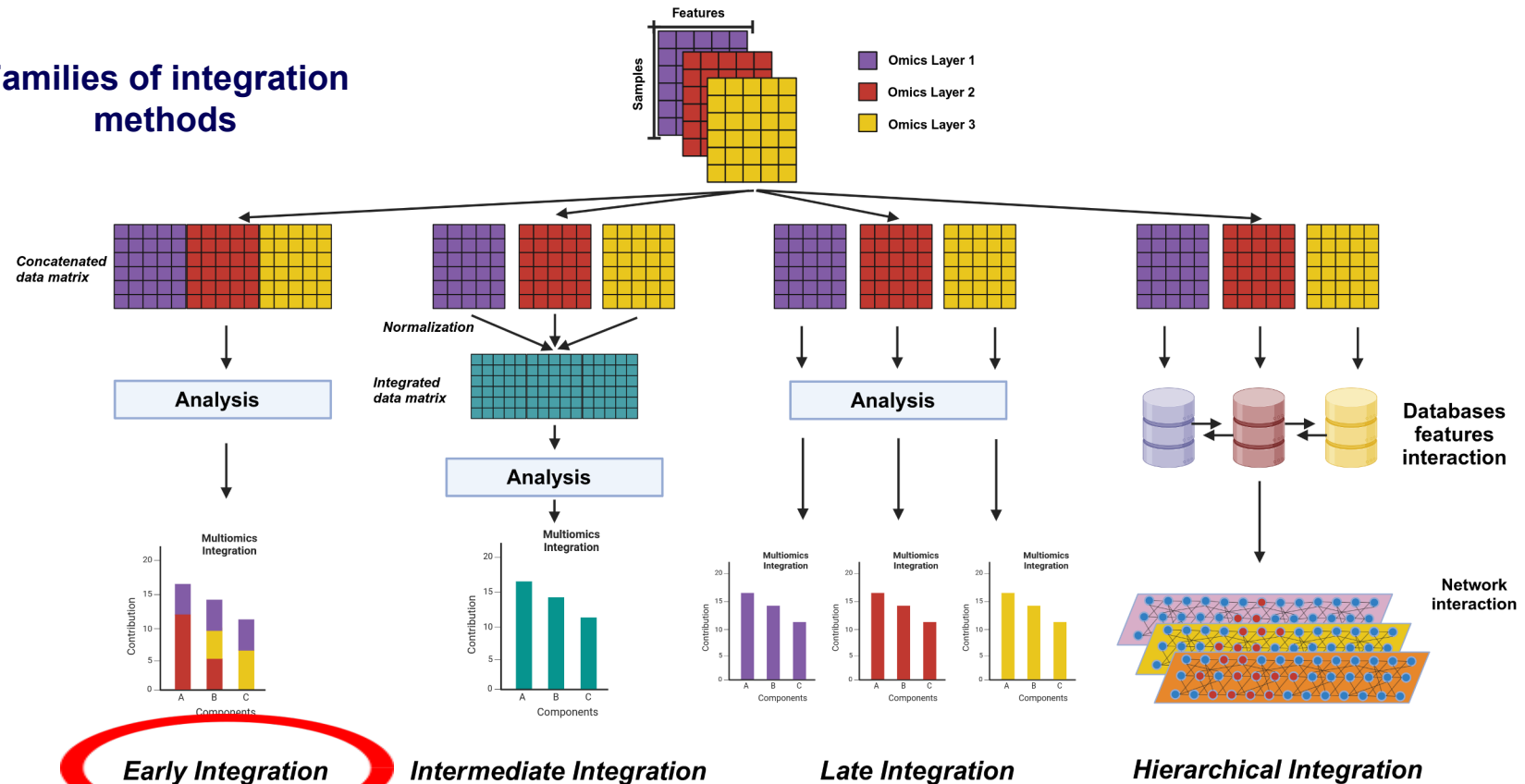
Multi-omics data integration, the grail of biology understanding?





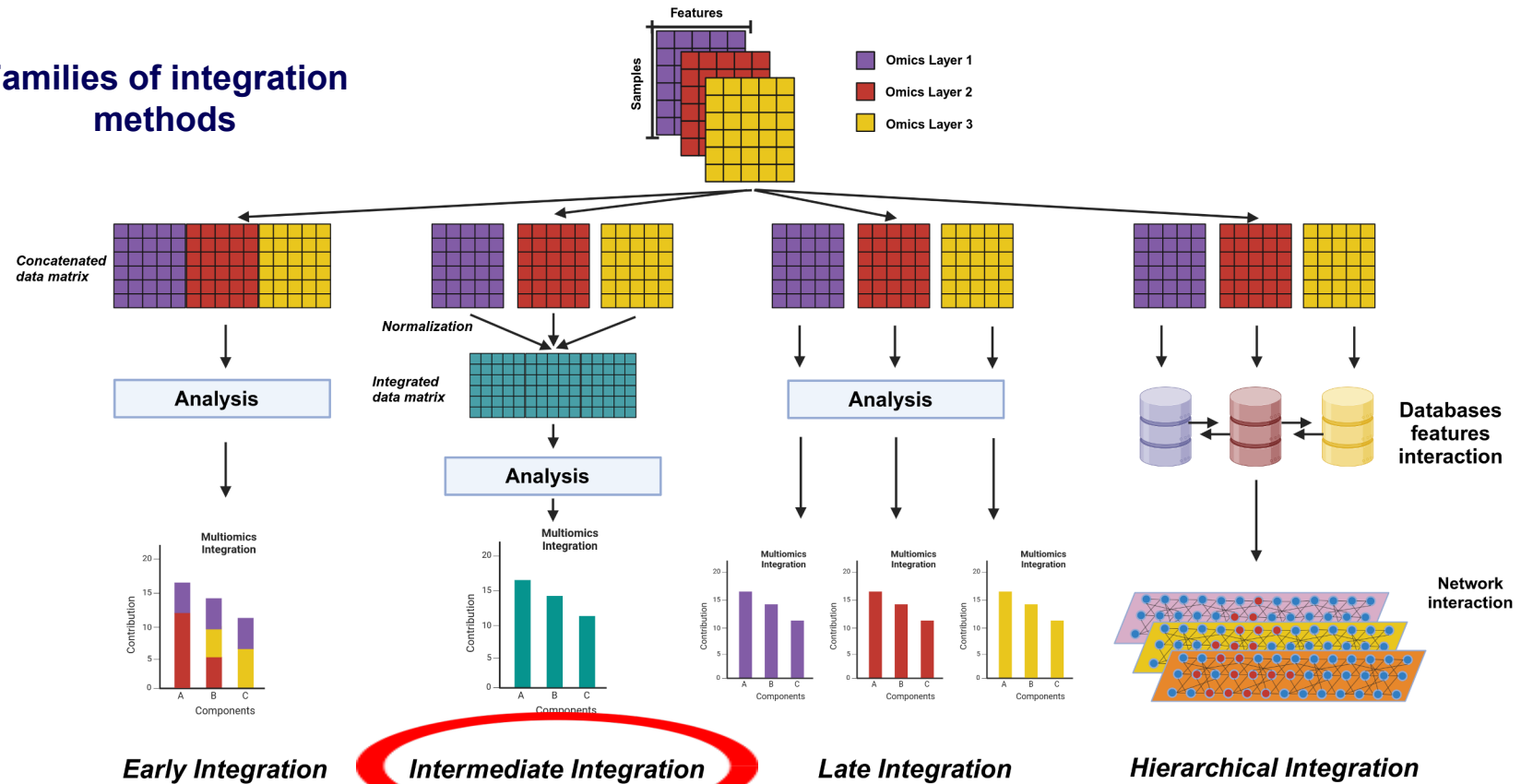


Families of integration methods



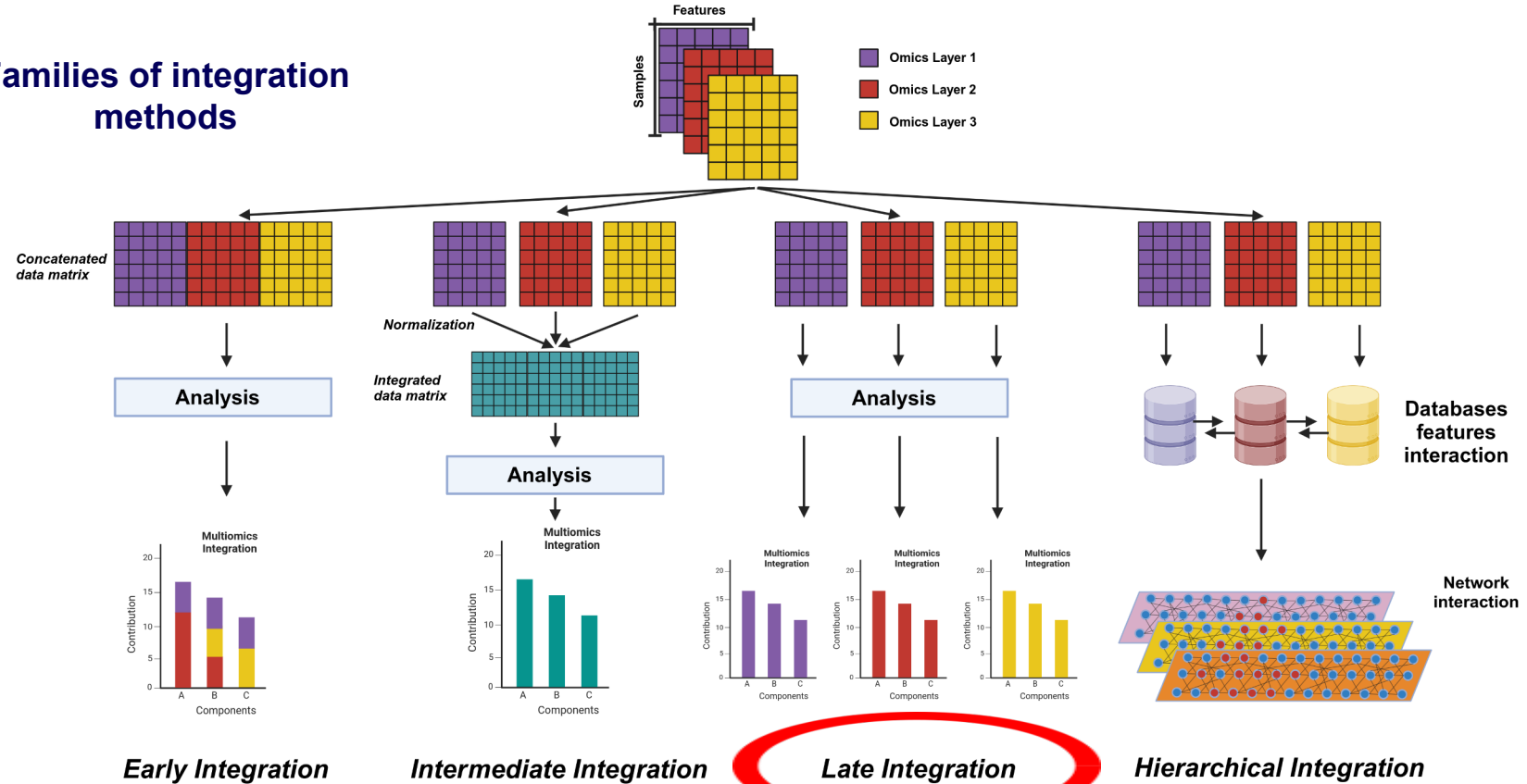


Families of integration methods



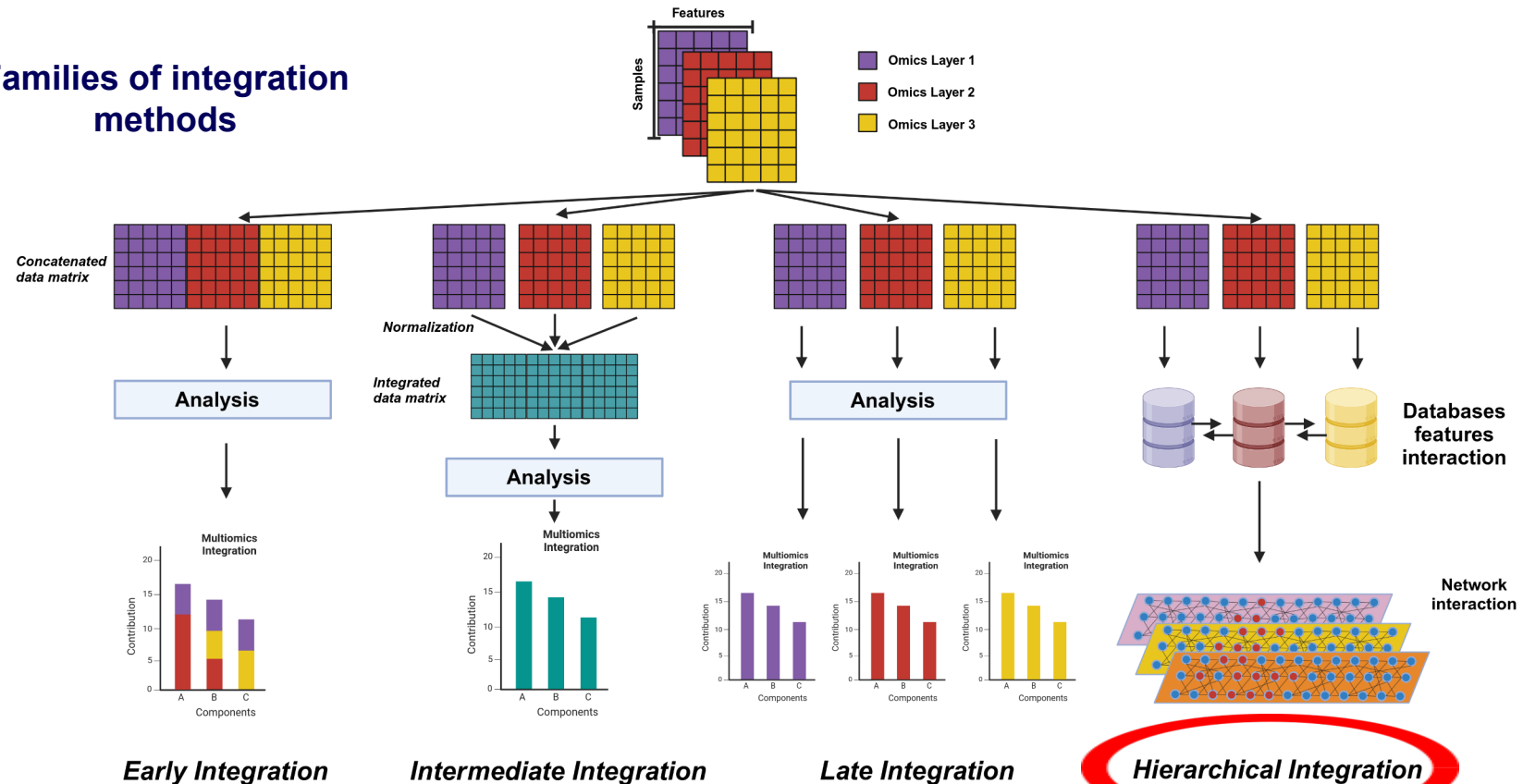


Families of integration methods





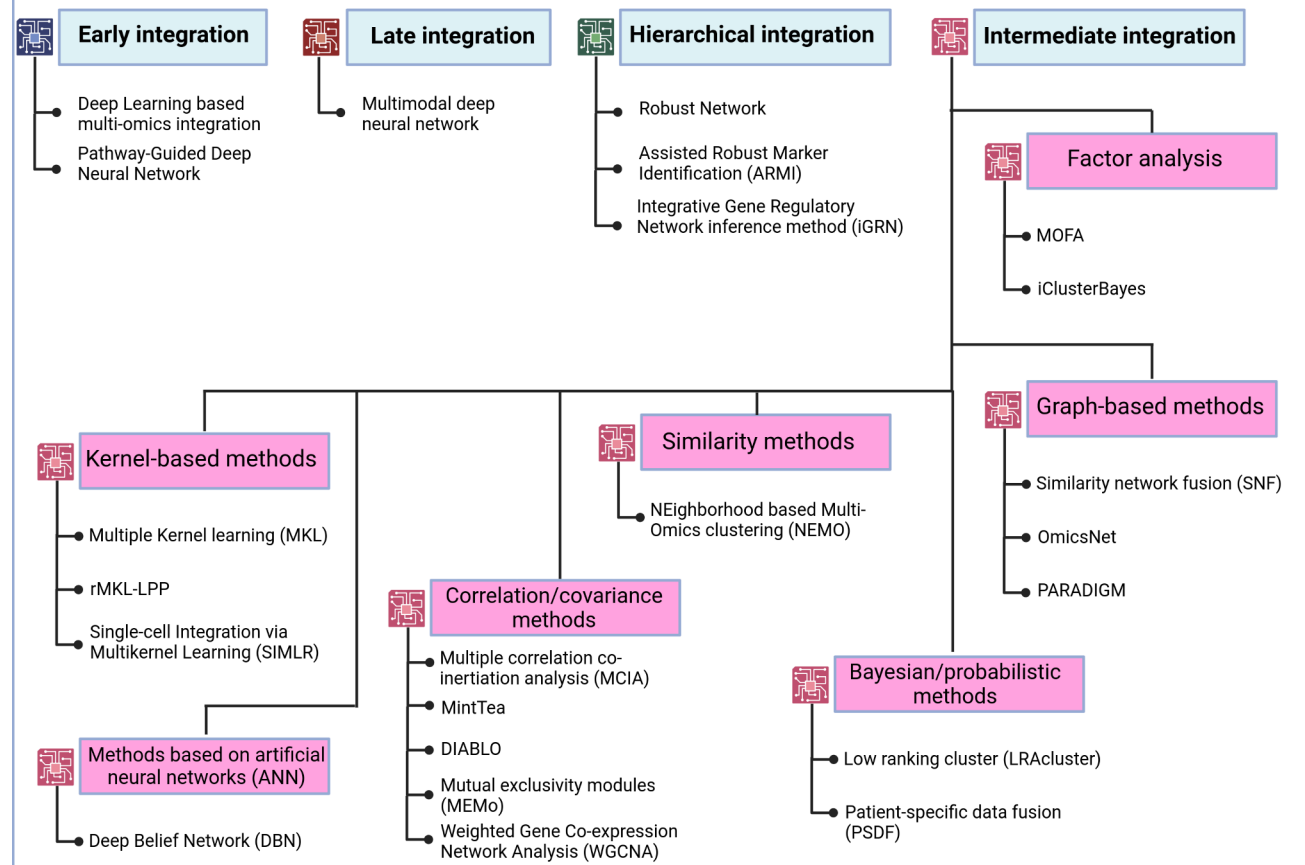
Families of integration methods





Intermediate Integration

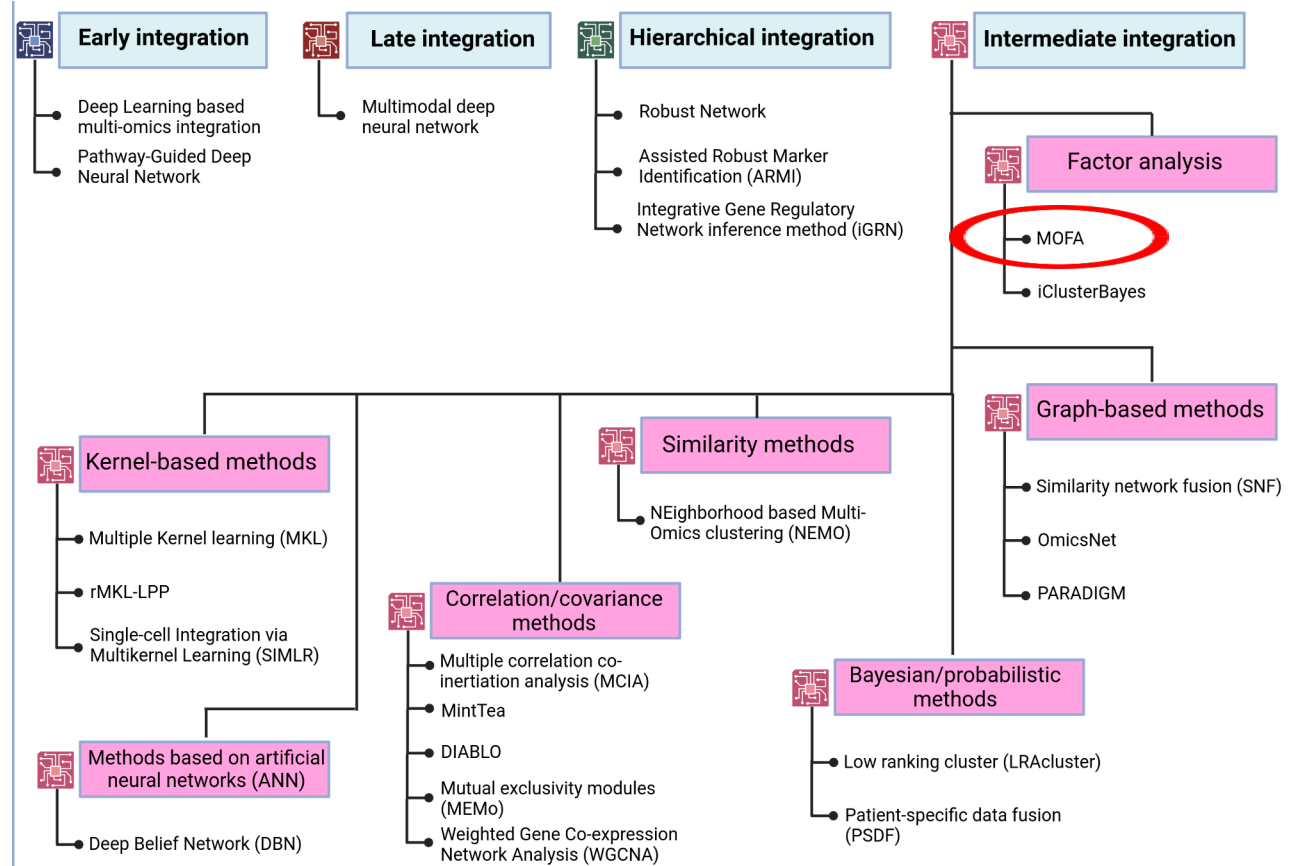
- Attempt to integrate data taking into consideration **size** and **distribution**
- Complexity of the normalization process





MOFA Integration

- Data **integration suited for Negative Binomial, Gaussian and Bernoulli distribution**
- Missing omics data handled
- Easy MOFA factor interpretation
- Low performance in non-linear data
- Arbitrary choice of number of factors

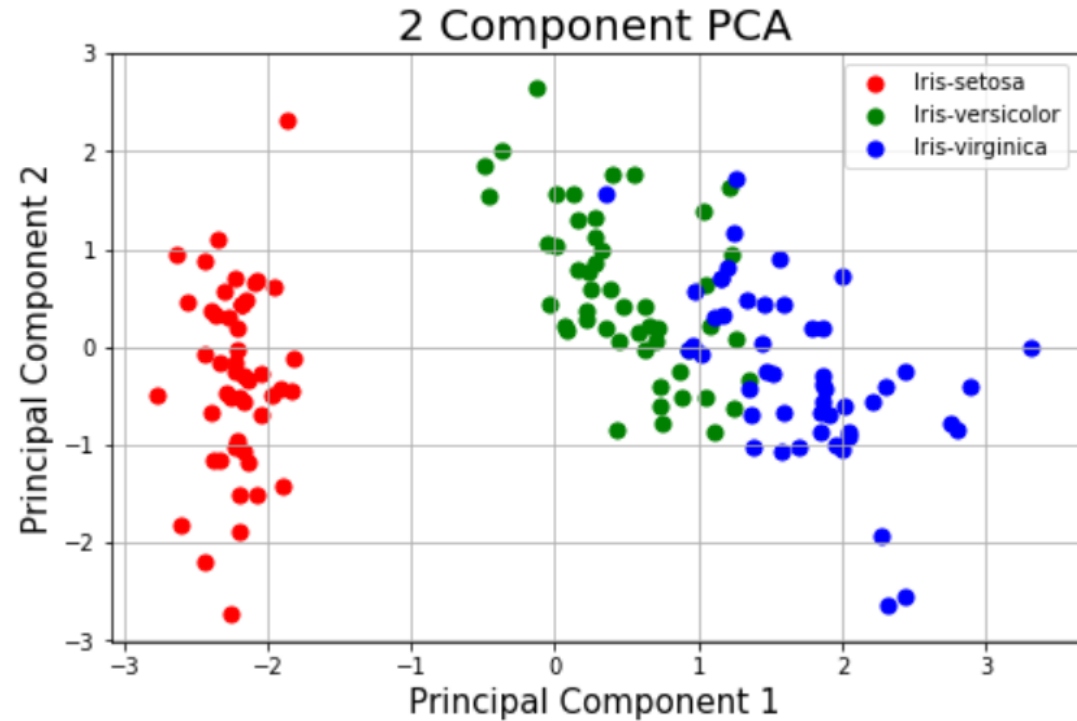
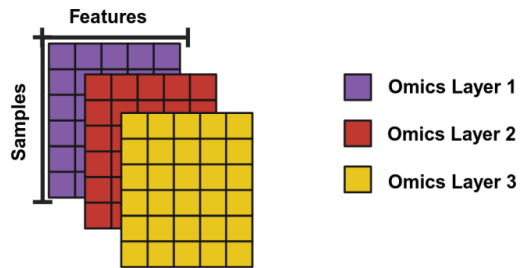




To make it simple

- MOFA is a dimensional reduction like PCA !

But more complicated
(bayesian model) and
including multiple omics

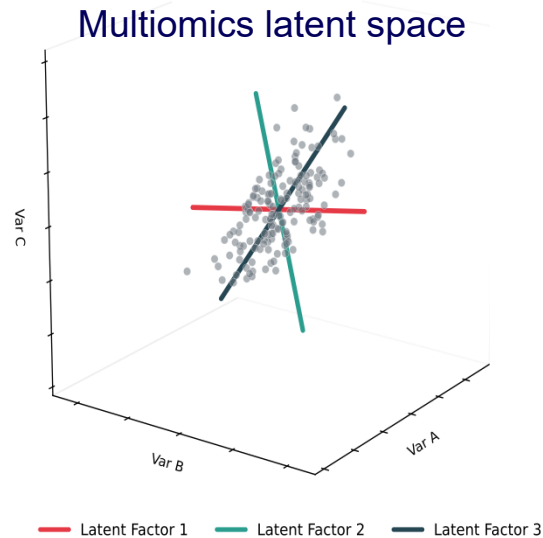
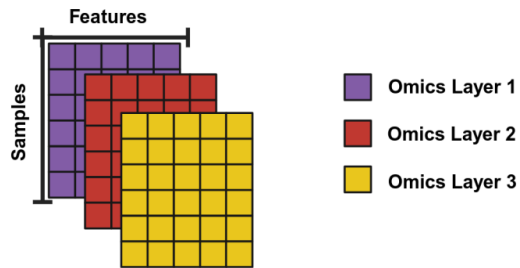




To make it simple

- MOFA is a dimensional reduction like PCA !

But more complicated
(bayesian model) and
including multiple omics



MOFA select the factors
with balanced weight
from all the omics.



INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☒ MOFA
☐ Diablo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS

N° factors
(0 = auto)

Factor to explore

Omics overlay

Main interface

⚙ Setup

Advanced Options

📊 QC & Imputation

JSON Preview

General

Metabolomics

MOFA & Diablo

SNF / NEMO

MintTea

Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations

Convergence mode

Contribution threshold

Advance option MOFA



INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☒ MOFA
☐ Diablo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS

N° factors
(0 = auto)

Factor to explore

Omics overlay

Number of factors to calculate. When 0 activates the automatic selection

Exclude samples with less than "" Omics shared

Setup

Advanced Options

QC & Imputation

JSON Preview

General

Metabolomics

MOFA & Diablo

SNF / NEMO

MintTea

Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations

Convergence mode

Contribution threshold

Converging speed, the slower the more accurate

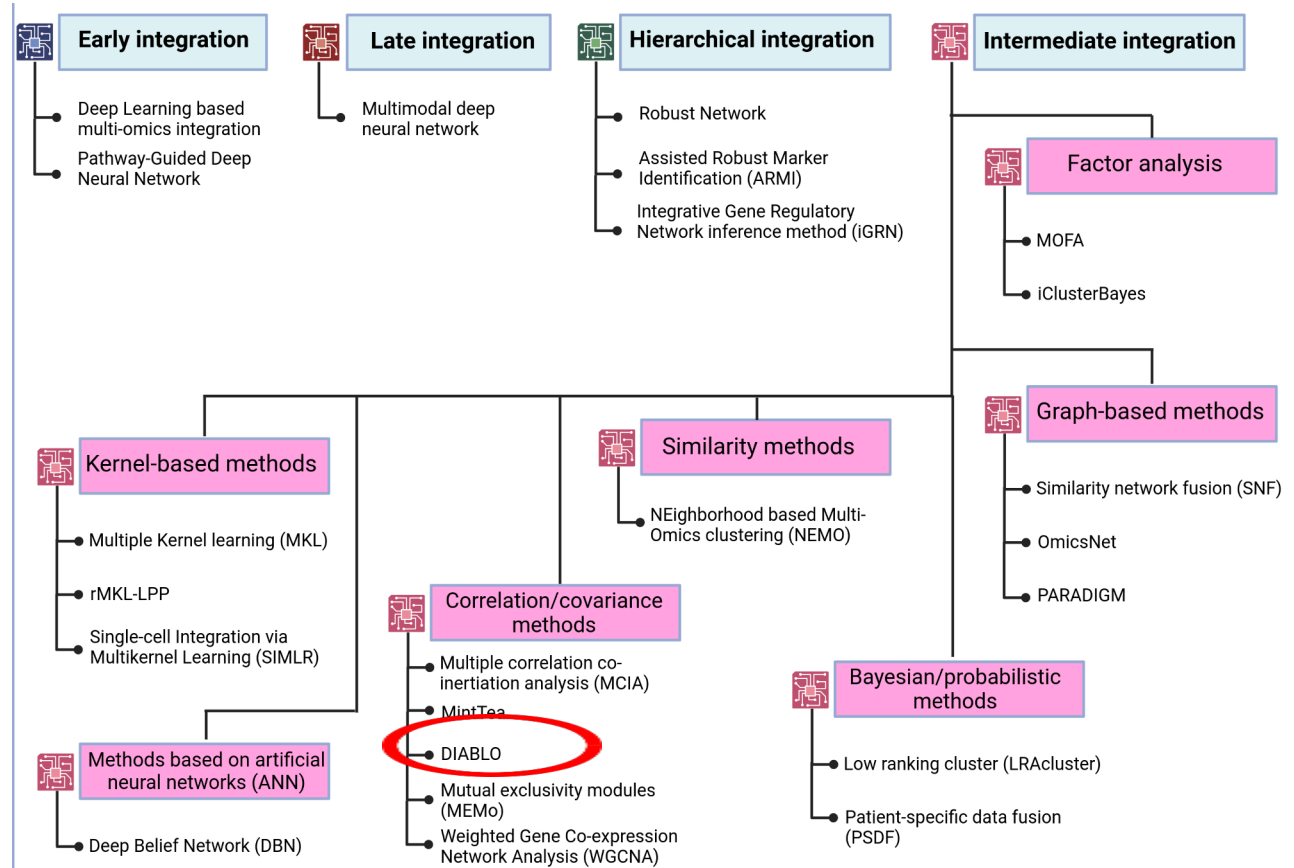
All features contribution to a factor are ranked from 0 to 1. Score threshold to filter features for interpretation analysis

Advance option MOFA



DIABLO Integration

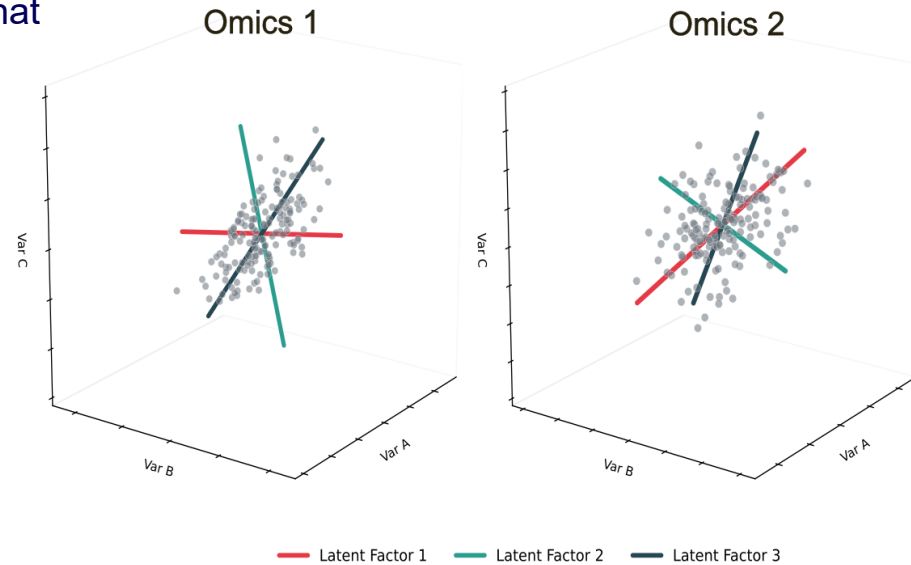
- Extension of the Partial least squares discriminant analysis (PLS-DA).
- Easy DIABLO factor interpretation
- Selection of features by LASSO-like selection
- Low performance in non-linear data
- Missing omics data non handled (imputation required!)





DIABLO Integration

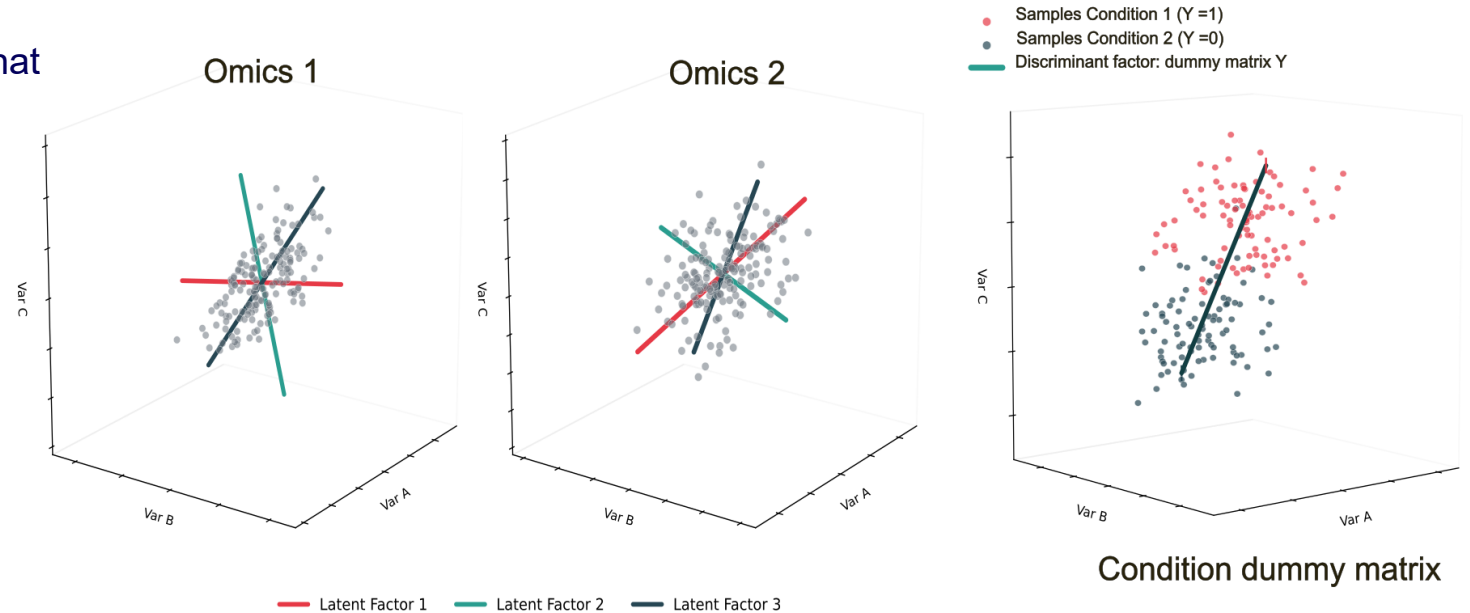
- Diablo select the factors that optimize the covariance among the latent factors calculated per omics





DIABLO Integration

- Diablo select the factors that optimize the covariance among the latent factors calculated per omics

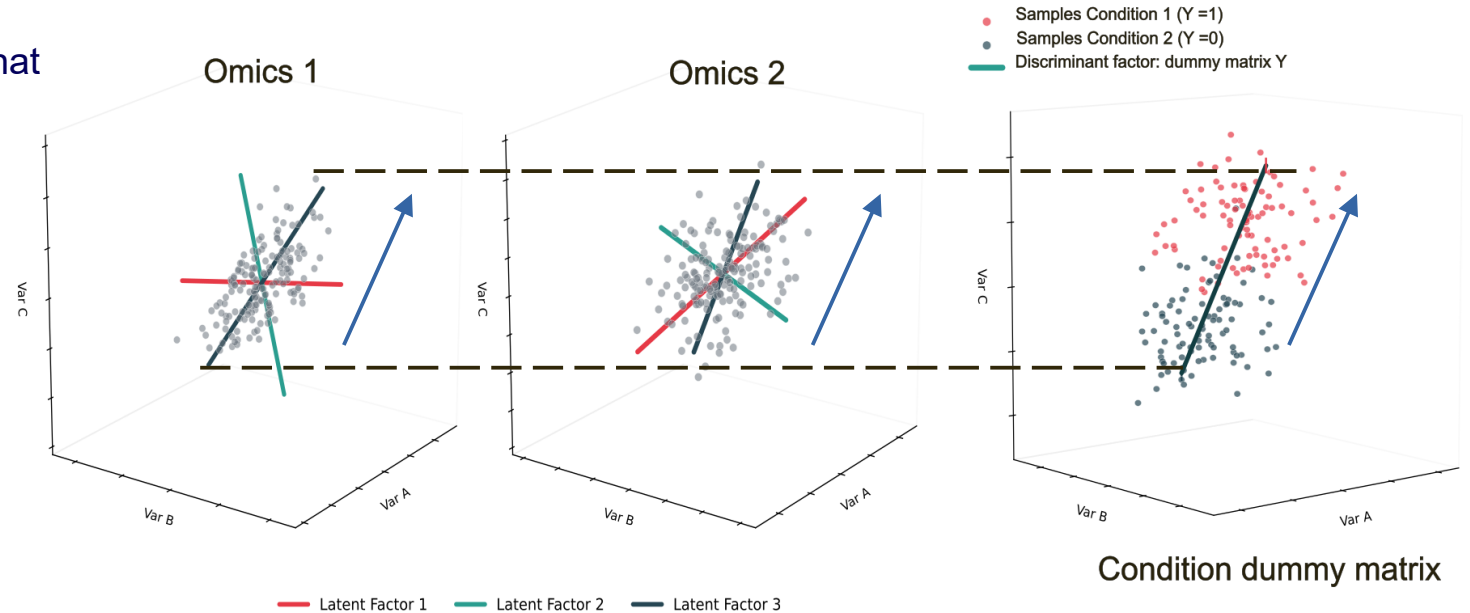


Dummy matrix provides the supervised frame to select the factors discriminating the two conditions



DIABLO Integration

- Diablo select the factors that optimize the covariance among the latent factors calculated per omics



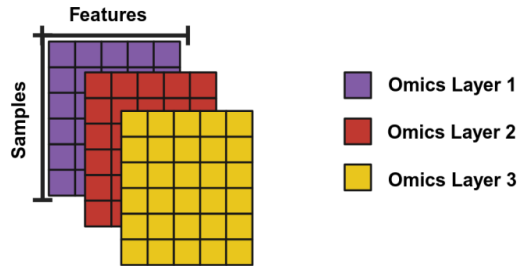
Dummy matrix provides the supervised frame to select the factors discriminating the two conditions



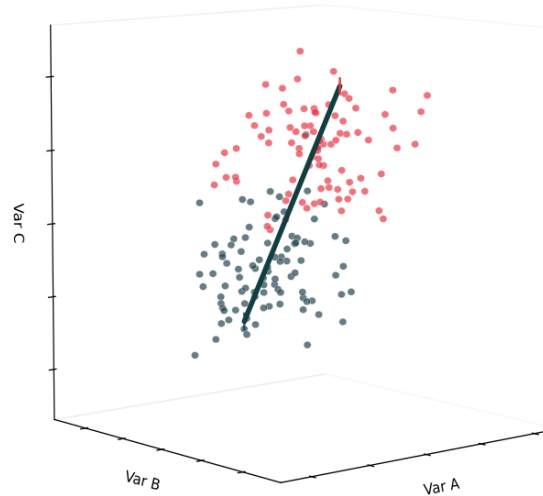
To make it simple

- DIABLO is a dimensional reduction like Linear discriminant analysis (LDA) !

But more complicated
(multi PLS-DA) and



● Samples Condition 1 ($Y=1$)
● Samples Condition 2 ($Y=0$)
— Discriminant factor: dummy matrix Y



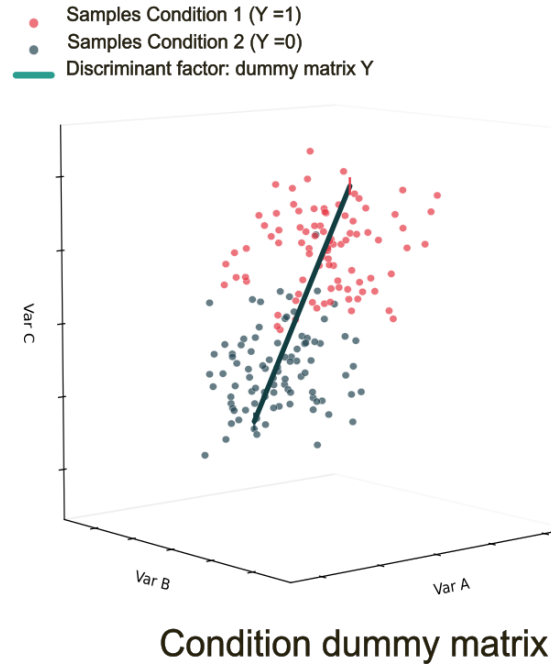
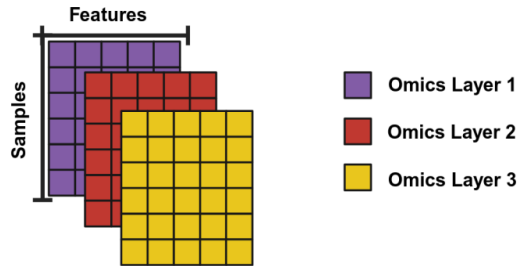
Condition dummy matrix



To make it simple

- DIABLO is a dimensional reduction like Linear discriminant analysis (LDA) !

But more complicated
(multi PLS-DA) and



Design matrix →
Define how much we
expect each omics to
correlate.

Perfectly = 1
Not at all = 0

	RNA	Methylome
RNA	1	0.8
Methylome	0.8	1



INTEGRATION OPTIONS

INTEGRATION
☒ Yes ☐ No

METHOD
☐ MOFA
☒ Diabolo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS
N° factors (0 = auto)

Factor to explore

Omics overlay

Number of factors to calculate. When 0 activates the automatic selection

Exclude samples with less than "" Omics shared

Main interface DIABLO

⊙ Setup **Advanced Options** QC & Imputation JSON Preview

General Metabolomics **MOFA & Diabolo** SNF / NEMO MintTea Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations

Convergence mode

Contribution threshold

DIABLO PARAMETERS

Feature selection

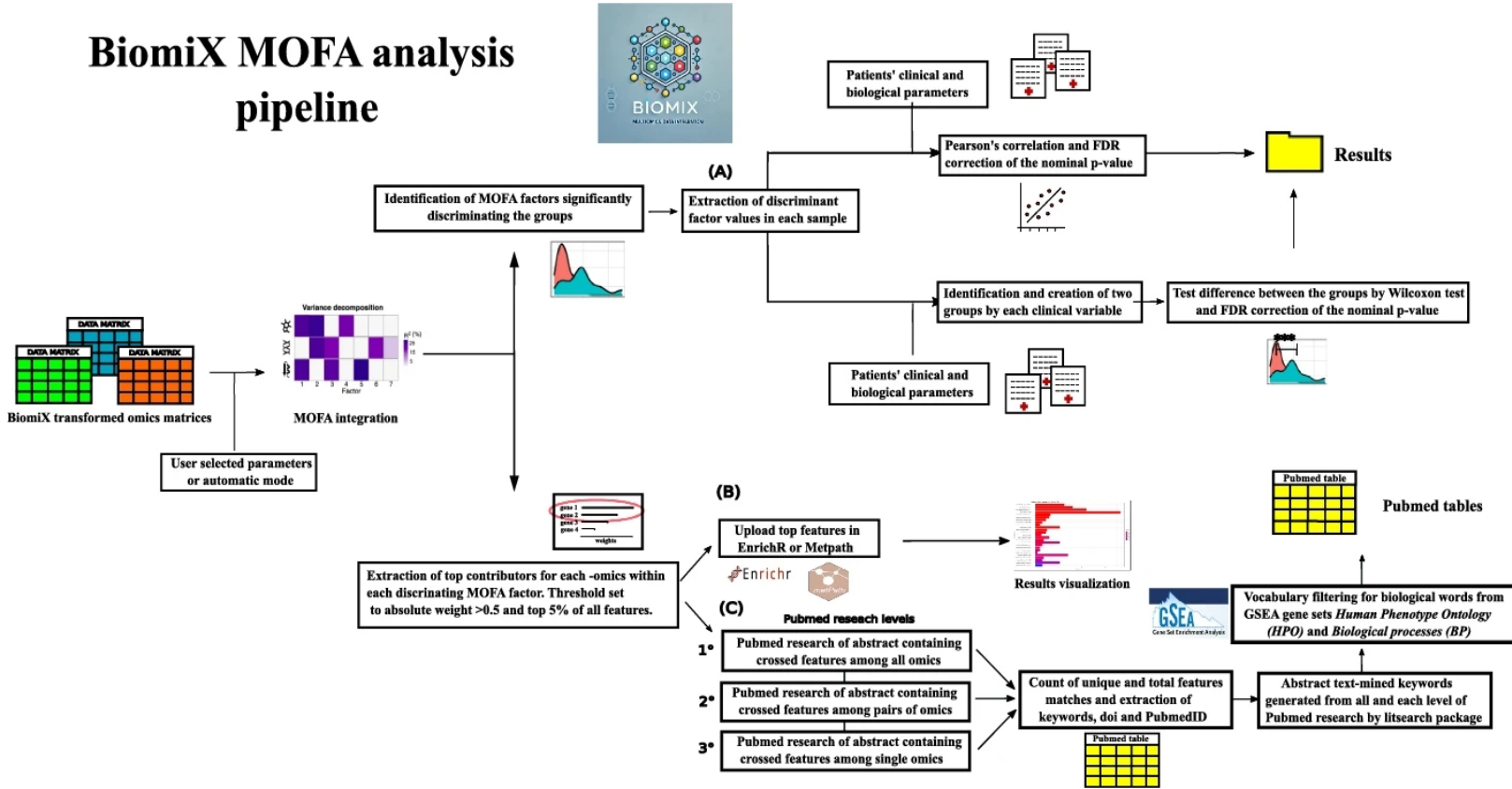
N° iterations

Design matrix file (optional)
 Nessun file (opzionale)

Advance option DIABLO



BiomiX MOFA analysis pipeline



What's new with BiomiX?

- Automatic selection MOFA/ DIABLO factors (stopping when new variance capture < 1%)
- Selection of factors separating the two conditions (semi-supervised)
- Automatic annotation of the latent factors
 - Correlation by clinical data
 - Pathway analysis
 - Bibliographic research



Setup **Advanced Options** QC & Imputation JSON Preview

General Metabolomics **MOFA & Diablo** SNF / NEMO MintTea Metadata Filtering

MOFA+ ALGORITHM PARAMETERS

Max iterations

5000

Convergence mode

fast

Contribution threshold

0,5

DIABLO PARAMETERS

Feature selection

NO

N° iterations

99

Design matrix file (optional)

Browse...

Nessun file (opzionale)

FACTOR INTERPRETATION – USED BY BOTH MOFA AND DIABLO

BIBLIOGRAPHY

Search type

Title/Abstract

N° articles retrieved

10

N° top Keywords extracted

100

N° top contributors

10

PATHWAY ANALYSIS

P.adj threshold

0,05

N° pathways shown

10

CLINICAL DATA INTEGRATION

Data types to correlate

☐ Numerical metadata

☐ Binary metadata

Numerical clinical data file (optional)

Browse...

No file (optional)

Binary clinical data file (optional)

Browse...

No file (optional)

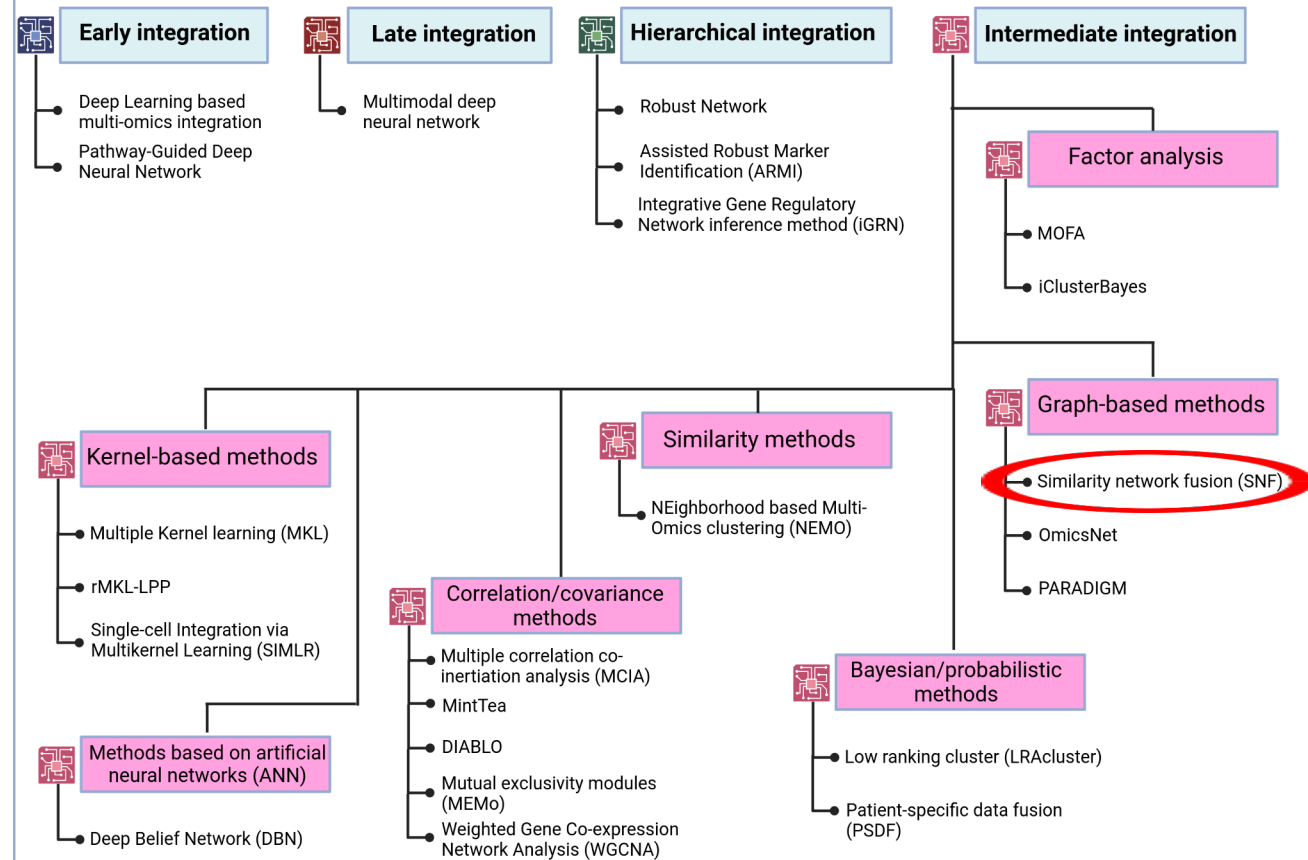
Advance option
MOFA/DIABLO



Similarity Network Fusion (SNF) Integration

(Wang et al, Nature Methods, 2014)

- unsupervised subtype discovery
- high-dimensional features into an integrated sample-sample network
- capture local/nonlinear relationships
- Network interpretation
- Limited biomarker interpretation
- Arbitrary choice of hyperparameters
- No missing-data handling
- Poor scaling with number of samples





How SNF works

Omic-specific similarity matrix using “**scaled exponential Euclidean kernel**”:

$$W(i, j) = \exp\left(-\frac{\rho^2(x_i, x_j)}{\mu \varepsilon_{i, j}}\right) \quad \varepsilon_{i, j} = \frac{\text{mean}(\rho(x_i, N_i)) + \text{mean}(\rho(x_j, N_j)) + \rho(x_i, x_j)}{3}$$

Computation of unimodal **global similarity matrix P**:

$$P(i, j) = \begin{cases} \frac{W(i, j)}{2 \sum_{k \neq i} W(i, k)} & , \text{ if } j \neq i \\ 1/2 & , \text{ if } j = i \end{cases}$$

Computation of the unimodal **local similarity matrix S**:

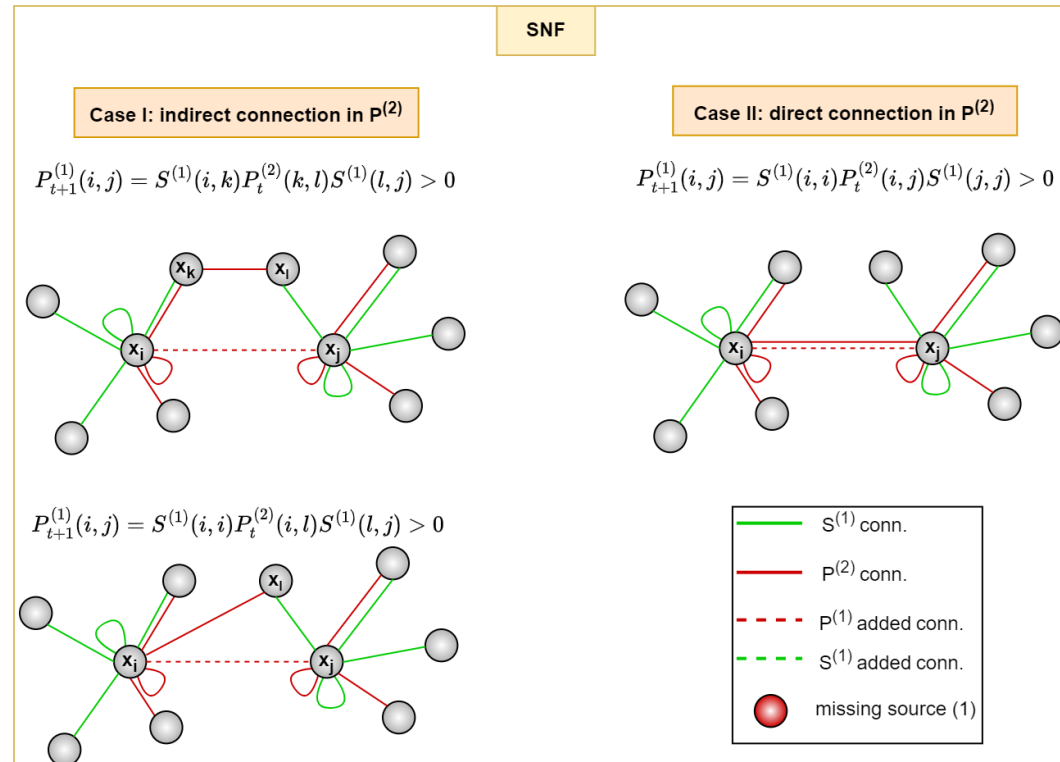
$$S(i, j) = \begin{cases} \frac{W(i, j)}{\sum_{k \in N_i} W(i, k)} & , \text{ if } j \in N_i \\ 0 & , \text{ otherwise} \end{cases}$$

where $N_i = \{x_k | x_k \in kNN(x_i) \cup \{x_i\}\}$.

Network integration through an **iterative parallel diffusion process and average** (two views $s=\{1,2\}$):

$$\begin{aligned} P_{t+1}^{(1)} &= S^{(1)} \times P_t^{(2)} \times S^{(1)T} \\ P_{t+1}^{(2)} &= S^{(2)} \times P_t^{(1)} \times S^{(2)T} \end{aligned} \quad \rightarrow \quad P_{t+1}^{(1)}(i, j) = \sum_{k \in N_i} \sum_{l \in N_j} S^{(1)}(i, k) P_t^{(2)}(k, l) S^{(1)}(l, j)$$

Spectral clustering for subgroups discovery.





Advanced Options SNF

Model:

- Number of nearest neighbours
- Variance (μ in similarity computation)
- Number of message-passing iterations

Interpretation:

- Column names for enrichment analysis
- Column names for survival analysis
- Condition column name (ground truth)

Clustering and features:

- Number cluster 2:X to test
- Number of variables to show in the heatmap
- Number features selected by variance
- Scaling (z-score)

Setup Advanced Options QC & Imputation JSON Preview

General Metabolomics MOFA & Diabolo **SNF / NEMO** MintTea Metadata Filtering

SNF PARAMETERS

N° neighbors: 5 Variance affinity matrix: 0,5 N° iterations: 10

NEMO PARAMETERS

N° neighbors: 5 Only common samples: TRUE

METADATA VARIABLES FOR SNF / NEMO

Columns for Variable Enrichment test: Columns for Time (add .time) and survival column (add .event): Condition column:

CLUSTERING AND FEATURE OPTIONS

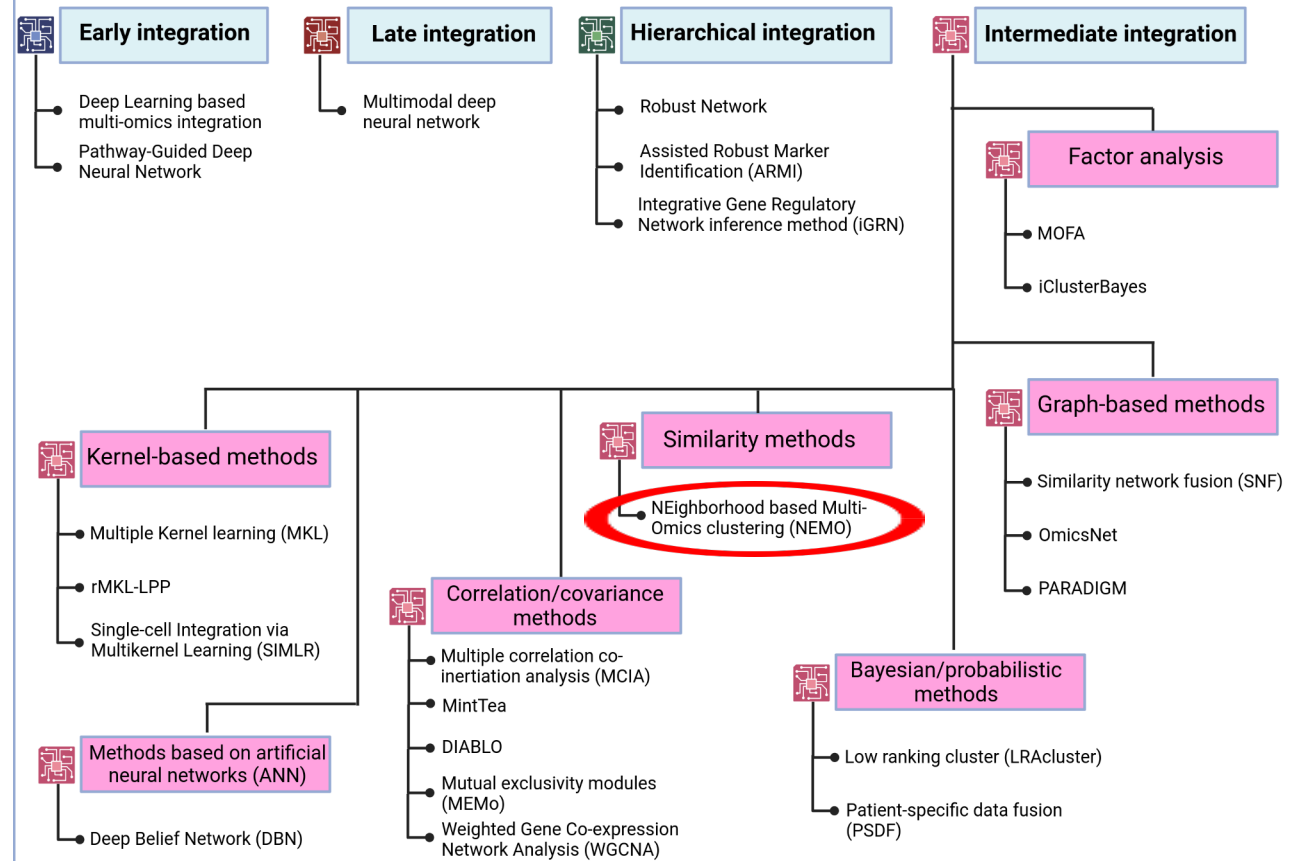
Max cluster range: 4 N° variables in heatmap: 20 Max N° features SNF / NEMO: 5000 Scaling the SNF/NEMO data: TRUE



NEighborhood based Multi-Omics clustering (NEMO) Integration

(Rappoport and Shamir, Bioinformatics, 2019)

- unsupervised subtype discovery
- high-dimensional features into an integrated sample-sample network
- capture local/nonlinear relationships
- Network interpretation
- Handles partial multi-omics datasets
- Limited biomarker interpretation
- Arbitrary choice of hyperparameters
- Assumes each pair of samples shares at least one measured omic
- Poor scaling with number of samples





How NEMO works

Omic-specific similarity matrix using “**scaled exponential Euclidean kernel**” (same as SNF).

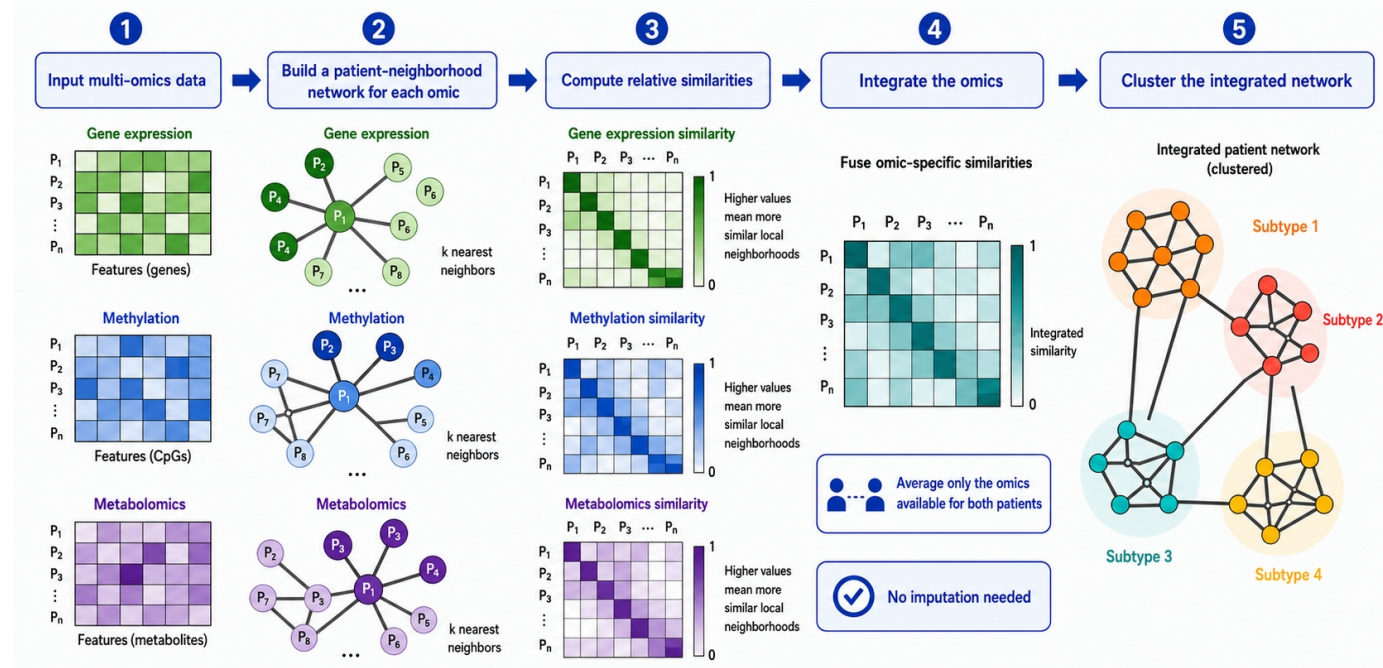
Define a **relative similarity matrix** for each omic l :

$$RS_l(i, j) = \frac{S_l(i, j)}{\sum_{r \in \eta_{li}} S_l(i, r)} \cdot I(j \in \eta_{li}) + \frac{S_l(i, j)}{\sum_{r \in \eta_{lj}} S_l(r, j)} \cdot I(i \in \eta_{lj})$$

Integration by **average** considering omics types available for both samples $JM(i, j)$:

$$ARS(i, j) = \frac{1}{|JM(i, j)|} \sum_{l \in JM(i, j)} RS_l(i, j)$$

Spectral clustering for subgroup discovery.





Setup

Advanced Options

QC & Imputation

JSON Preview

General

Metabolomics

MOFA & Diablob

SNF / NEMO

MintTea

Metadata Filtering

SNF PARAMETERS

N° neighbors

5

Variance affinity matrix

0,5

N° iterations

10

NEMO PARAMETERS

N° neighbors

5

Only common samples

TRUE

METADATA VARIABLES FOR SNF / NEMO

Columns for Variable Enrichment test

Columns for Time (add .time) and survival column (add .event)

Condition column

CLUSTERING AND FEATURE OPTIONS

Max cluster range

4

N° variables in heatmap

20

Max N° features SNF / NEMO

5000

Scaling the SNF/NEMO data

TRUE

Advanced Options NEMO

Model:

- Number of nearest neighbours
- Consider only samples having all omics available

Interpretation:

- Column names for enrichment analysis
- Column names for survival analysis
- Condition column name (ground truth)

Clustering and features:

- Number cluster 2:X to test
- Number of variables to show in the heatmap
- Number features selected by variance
- Scaling (z-score)

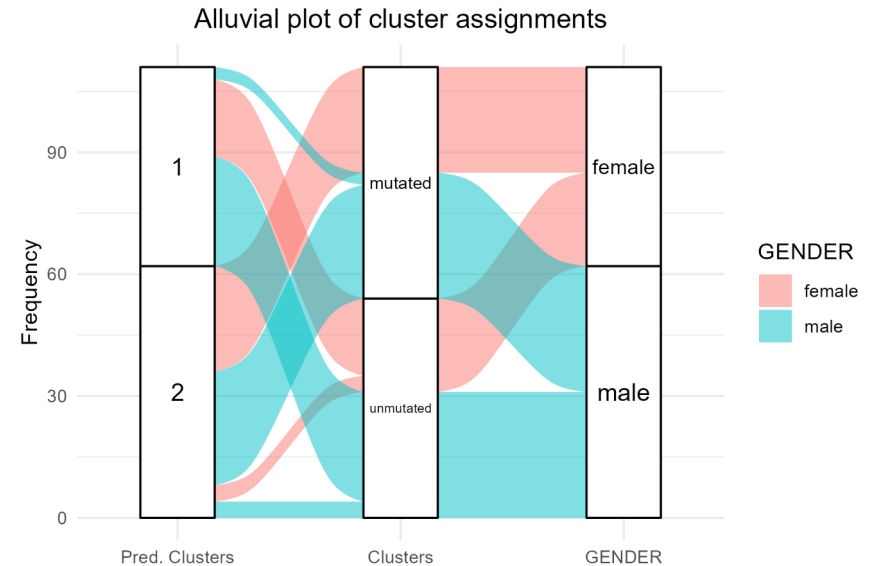


Number of clusters optimization SNF/NEMO

- Optimal number of clusters estimation: **eigengap/modified eigegap method**
- Internal clustering **validation metrics** (i.e. "Mean Silhouette Coefficient", "Dunn Index", "Entropy" and "Calinski and Harabasz index")

Implemented interpretation methods SNF/NEMO

- Similarities heatmap for single omics and integrated matrix
- Network (graph) for single omics and integrated network
- External quality indices (i.e. ARI, AMI, NMI, NVI) if CONDITION (ground truth available)
- Feature importance based on NMI
- Variables enrichment (chi-squared test for categorical, Kruskal-Wallis for continuous variables)
- Survival analysis (log-rank test and Kaplan-Meier plots)
- Alluvial plot comparing clustering, CONDITION (if any), variables of interest







Biomix v2.5

Multiomics analysis pipeline



1. METADATA FILE

File must contain a 'CONDITION' column (.xlsx, .tsv, .csv)

Select metadata file...

EGAS00001001746_metadata_CLL.xls

184 samples · 3 conditions: mutated, unmutated, unknown

2. COMPARISON GROUPS

Condition (GROUP_1)

mutated

Control (GROUP_2)

unmutated

3. OUTPUT FOLDER

Folder where Biomix will save results and the JSON configuration file

Select folder...

C:/Users/crist/Desktop/Biomix2.5

- 1) Load EGAS00001001746_metadata_CLL.xls from Metadata folder.
- 2) Set the mutated vs unmutated comparison
- 3) Output folder in Biomix 3.0 folder



OMICS INPUTS

INPUT	PREVIEW QC	SINGLE OMICS ANALYSIS	DATA TYPE	INTEGRATION	LABEL	FILTER BY LABEL	DATA UPLOAD
Input 1	☐	☐	Transcriptomics	☒	RNA	☐	Browse file... EGAS00001001746_transcriptomics.xls C:/Users/crist/Desktop/BioMiX2.5/Transcriptomics/INPUT/EGAS00001001746_transcriptomics.xls
Input 2	☐	☐	Methylomics	☒	Methy	☐	Browse file... EGAS00001001746_methylomics.xls C:/Users/crist/Desktop/BioMiX2.5/Methylomics/INPUT/EGAS00001001746_methylomics.xls
Input 3	☐	☐	Undefined	☐	Label input 3	☐	Browse file... No file selected

- 4) Load EGAS00001001746_transcriptomics.xls from Transcriptomics folder.
- 5) Load EGAS00001001746_methylomics.xls from Methylomics folder.
- 6) Click on integration to select the slot to integrate.



INTEGRATION OPTIONS

INTEGRATION

☒ Yes ☐ No

METHOD

- ☒ MOFA
☐ Diabolo
☐ SNF
☐ NEMO
☐ MintTea

PARAMETERS

N° factors
(0 = auto)

10

Factor to explore

1

Omics overlay

0

Select 10 factors

⚙️ Advanced Options

▶ Run Biomix

📄 Download JSON

✅ Ready to run





1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors

Each omics has the positive and negative contributors for the “discriminating” factors

Example:

	A	B	C
1	weights	features	abs_weight
2	-0.50266	cg07825294/DAZAP1	0.502661
3	-0.50448	cg07499142/PIK3CD	0.504481
4	-0.50682	cg01320698/PIK3CD	0.506823
5	-0.51647	cg03437605/RHOBTB	0.516468
6	-0.52646	cg11978120/NA	0.526455
7	-0.52949	cg14999716/HCF2	0.529491
8	-0.53092	cg24545687/EP400	0.53092
9	-0.53151	cg09416521/FOXK2	0.53151

Factors_Associated_Articles	20/05/2026 08:48	Cartella di file	
Distribution_factors_contributions_mutat...	20/05/2026 08:56	Adobe Acrobat D...	24 KB
Methylomics_Methylomics_neg_factor_1.t...	20/05/2026 08:41	File TSV	13 KB
Methylomics_Methylomics_neg_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_neg_factor_6.t...	20/05/2026 08:41	File TSV	12 KB
Methylomics_Methylomics_neg_factor_10...	20/05/2026 08:42	File TSV	1 KB
Methylomics_Methylomics_pos_factor_1.t...	20/05/2026 08:41	File TSV	2 KB
Methylomics_Methylomics_pos_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_pos_factor_6.t...	20/05/2026 08:41	File TSV	1 KB
Methylomics_Methylomics_pos_factor_10...	20/05/2026 08:42	File TSV	1 KB
MOFA_FACTORS_WEIGHTS.tsv	20/05/2026 08:42	File TSV	31 KB
MOFA_MATRIX_mutated_vs_unmutated_1...	20/05/2026 08:40	File TSV	1 KB
MOFA_mutated_vs_unmutated.pdf	20/05/2026 08:42	Adobe Acrobat D...	2.273 KB
MOFA_SEPARATION_mutated_vs_unmutat...	20/05/2026 08:40	File TSV	1 KB



1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors

Latent factor value (10 factors) per each sample. Dimensional reduction-like.

Variance explained by each factor within each omics

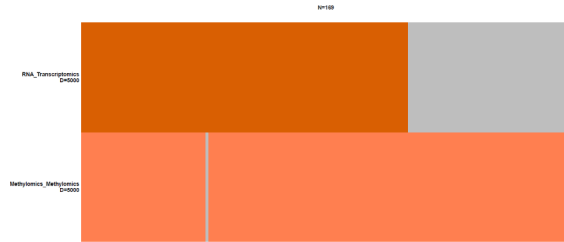
P.adjusted value of the “discrimination” power of each factor among the two conditions

Factors_Associated_Articles	20/05/2026 08:48	Cartella di file	
Distribution_factors_contributions_mutat...	20/05/2026 08:56	Adobe Acrobat D...	24 KB
Methylomics_Methylomics_neg_factor_1.t...	20/05/2026 08:41	File TSV	13 KB
Methylomics_Methylomics_neg_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_neg_factor_6.t...	20/05/2026 08:41	File TSV	12 KB
Methylomics_Methylomics_neg_factor_10...	20/05/2026 08:42	File TSV	1 KB
Methylomics_Methylomics_pos_factor_1.t...	20/05/2026 08:41	File TSV	2 KB
Methylomics_Methylomics_pos_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_pos_factor_6.t...	20/05/2026 08:41	File TSV	1 KB
Methylomics_Methylomics_pos_factor_10...	20/05/2026 08:42	File TSV	1 KB
MOFA_FACTORS_WEIGHTS.tsv	20/05/2026 08:42	File TSV	31 KB
MOFA_MATRIX_mutated_vs_unmutated_1...	20/05/2026 08:40	File TSV	1 KB
MOFA_mutated_vs_unmutated.pdf	20/05/2026 08:42	Adobe Acrobat D...	2.273 KB
MOFA_SEPARATION_mutated_vs_unmutat...	20/05/2026 08:40	File TSV	1 KB

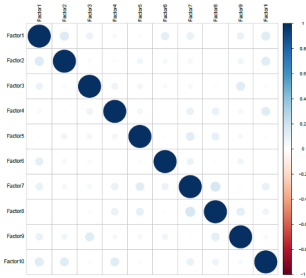


1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors

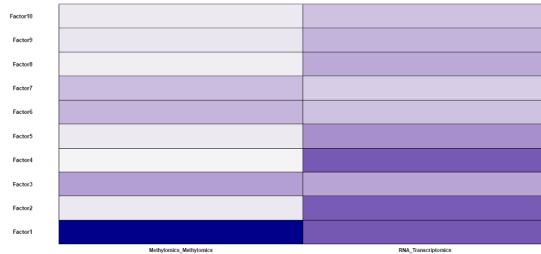
Main pdf results including



Sample distribution
per omics

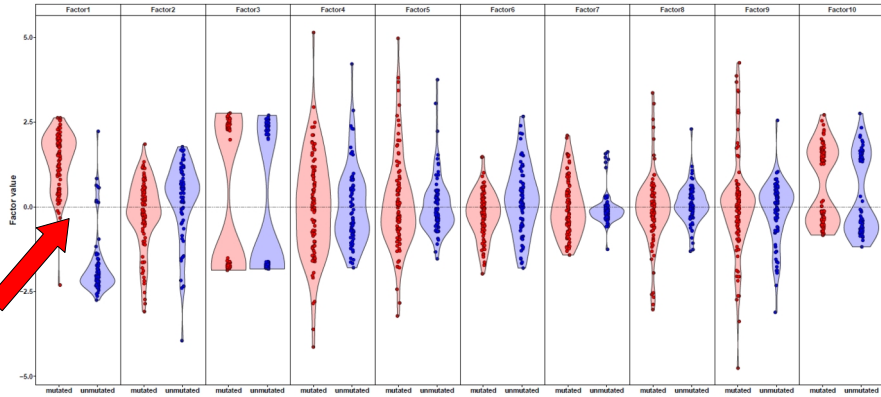


Factor co-
correlation test



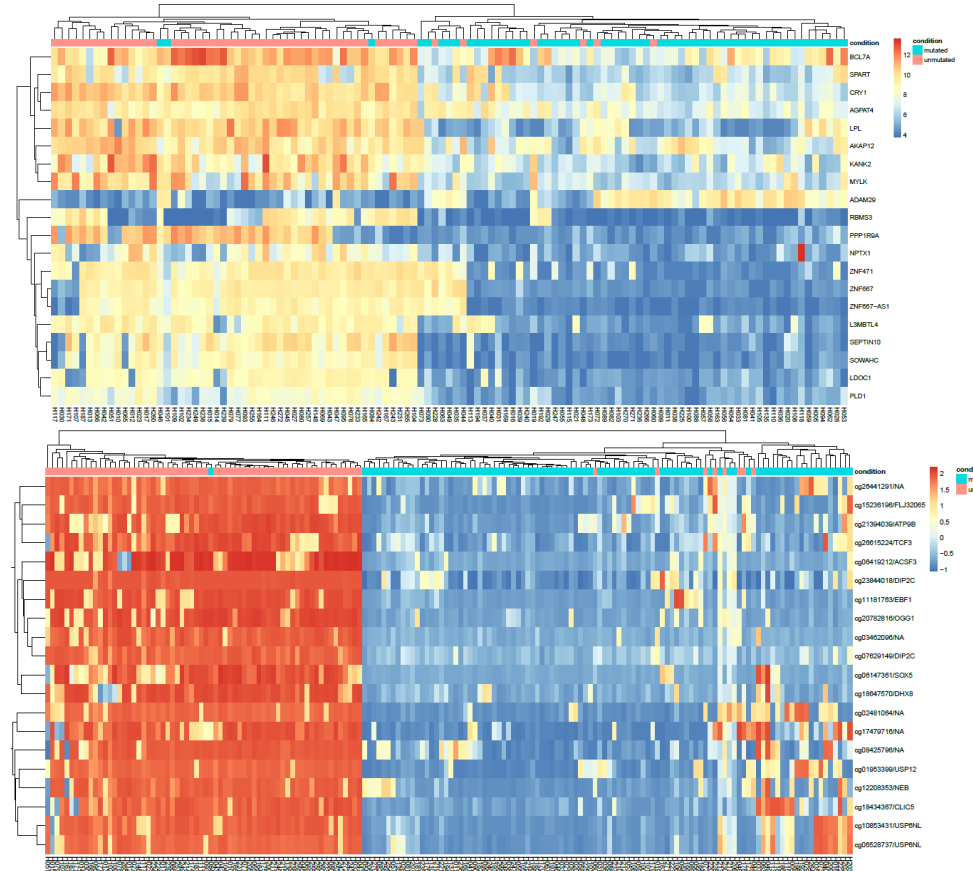
Variance explained per omics
and factor

Factors_Associated_Articles	20/05/2026 08:48	Cartella di file	
Distribution_factors_contributions_mutat...	20/05/2026 08:56	Adobe Acrobat D...	24 KB
Methylomics_Methylomics_neg_factor_1.t...	20/05/2026 08:41	File TSV	13 KB
Methylomics_Methylomics_neg_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_neg_factor_6.t...	20/05/2026 08:41	File TSV	12 KB
Methylomics_Methylomics_neg_factor_10...	20/05/2026 08:42	File TSV	1 KB
Methylomics_Methylomics_pos_factor_1.t...	20/05/2026 08:41	File TSV	2 KB
Methylomics_Methylomics_pos_factor_2.t...	20/05/2026 08:41	File TSV	3 KB
Methylomics_Methylomics_pos_factor_6.t...	20/05/2026 08:41	File TSV	1 KB
Methylomics_Methylomics_pos_factor_10...	20/05/2026 08:42	File TSV	1 KB
MOFA_FACTORS_WEIGHTS.tsv	20/05/2026 08:42	File TSV	31 KB
MOFA_MATRIX_mutated_vs_unmutated_1...	20/05/2026 08:40	File TSV	1 KB
MOFA_mutated_vs_unmutated.pdf	20/05/2026 08:42	Adobe Acrobat D...	2.273 KB
MOFA_SEPARATION_mutated_vs_unmutat...	20/05/2026 08:40	File TSV	1 KB



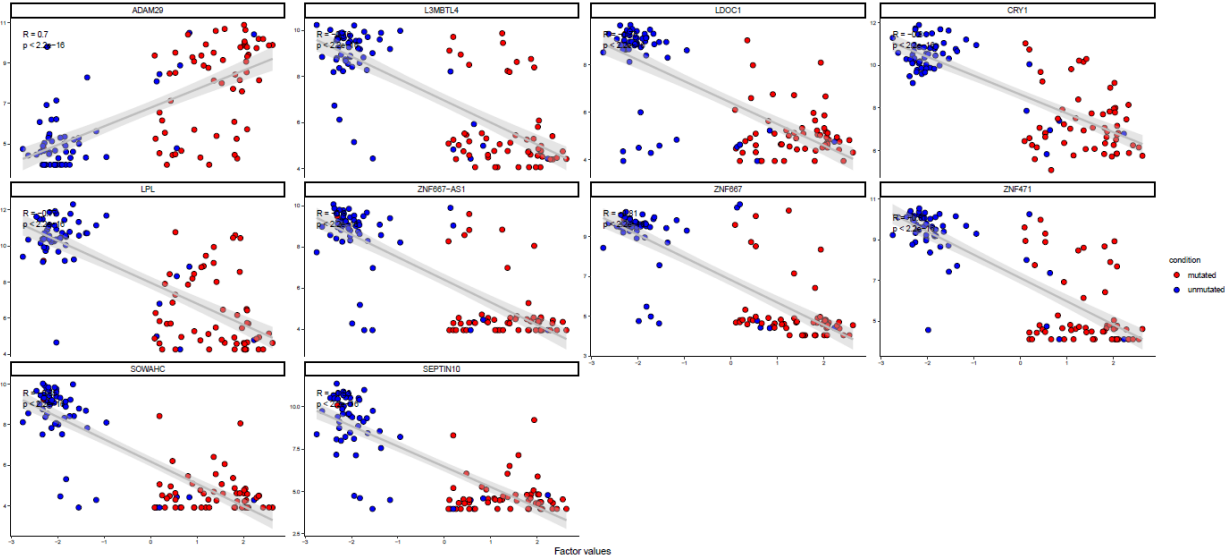
Violin plot factor separation per condition

Heatmap to features
factor per omics

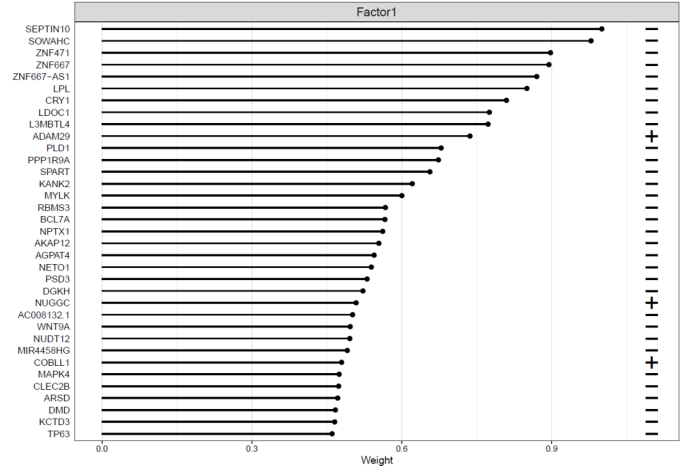




1) Open output folder: Integration\OUTPUT\MOFA_mutated_vs_unmutated_10_factors



Top contributor discriminatory power per condition



Top contributor weights.



Nice, but what is behind
factor 1?



3 Level of interpretation

- Pubmed research
- Pathway analysis
- Clinical correlation

🔍 FACTOR INTERPRETATION – USED BY BOTH MOFA AND DIABLO

BIBLIOGRAPHY

Search type:

N° articles retrieved:

N° top Keywords extracted:

N° top contributors:

PATHWAY ANALYSIS

P.adj threshold:

N° pathways shown:

CLINICAL DATA INTEGRATION

Data types to correlate:

- ☒ Numerical metadata
- ☒ Binary metadata

Numerical clinical data file (optional): [EGAS00001001746_clinical_numeric.xlsx](#)

Binary clinical data file (optional): [EGAS00001001746_clinical_binary.xls](#)



Nice, but what is behind factor 1?



3 Level of interpretation

- Pubmed research
- Pathway analysis
- Clinical correlation

FACTOR INTERPRETATION - USED BY BOTH MOFA AND DIABLO

BIBLIOGRAPHY

Search type: Title/Abstract (dropdown)
N° articles retrieved: 10 (dropdown)
N° top Keywords extracted: 100 (dropdown)
N° top contributors: 10 (text input)

PATHWAY ANALYSIS

P.adj threshold: 0,05 (text input)
N° pathways shown: 10 (text input)

CLINICAL DATA INTEGRATION

Data types to correlate:
☒ Numerical metadata
☒ Binary metadata

Numerical clinical data file (optional):
Browse... EGAS00001001746_clinical_numeric.xlsx

Binary clinical data file (optional):
Browse... EGAS00001001746_clinical_binary.xls

1) Re-run including Numeric and binary clinical data. These will be correlated and tested with the MOFA factors calculated.

Are they associated?



Open:BiomiX2.5\Clinical_data\OUTPUT\Clinical_mutated_vs_unmutated_10_factors



	A	B	C	D	E
1	clinic	means	pval	p.adj	
2	IGHV.Uppsala.U.M.1	3.108624	5.3918365	4.7448162020209e-23	
3	del11q22.3	-1.86591	4.7366701	2.08413488138209e-05	
4	BRAF	-1.7882	0.001813	0.053191	
5	TP53	-1.12497	0.002874	0.063222	
6	LAMA1	-2.38988	0.007647	0.134582	
7	KLHL6	2.10113	0.016083	0.235885	
8	del13q14_bi	0.956899	0.019169	0.240977	
9	del9p21.3	-1.61911	0.036384	0.329951	
10	del17p13	-1.06866	0.035695	0.329951	
11	MED12	-1.67928	0.037494	0.329951	

1) Conclusion:
Factor 1 is directly
associated to the IGHV
mutation measurement



To be included

- Include the same step to step slides for DIABLO, SNF and NEMO

To be included

- Use MOFA factor 1 as story driving, looking for it in other integration methods.



Part 2 – Multi-Omics Integration (2 h)

15:30 – 16:00 — Overview of multi-omics integration strategies: latent-factor and graph-based approaches (MOFA, DIABLO, SNF, NEMO) (30 min) Led by Jessica Gliozzo

Topic and Objective: This section provides an introduction to current integration methodologies (early, intermediate, and late integration), with a focus on intermediate integration methods. We will discuss the main challenges in multi-omics integration, including missing data, high-dimensionality and how selected methods (MOFA, DIABLO, SNF, NEMO) address these challenges, highlighting the pros and cons of each integration approach.

16:00 – 16:30 — MOFA and DIABLO: latent factor models and supervised integration (30 min) Led by Cristian Iperi

Topic and Objective: This section focuses on the MOFA and DIABLO algorithms, covering best practices and parameter selection. The CLL dataset (EGAS00001001746; doi: 10.1172/JCI93801). will be used to demonstrate their usage within BiomiX. The BiomiX implementation for model and factor selection will be presented, along with the pathway analysis, clinical correlation, and literature-based interpretation tools available within the software.

16:30 – 17:00 — SNF and NEMO: graph-based and similarity network approaches (30 min) Led by Jessica Gliozzo

Topic and Objective: This section focuses on the SNF and NEMO algorithms, covering best practices and parameter selection. The CLL dataset will again be used, allowing a direct comparison with the results obtained from MOFA and DIABLO. The BiomiX implementation for integration, clustering, graph visualization and results interpretation will be demonstrated.

17:00 – 17:30 — Final discussion and Q&A (30 min)

Topic and Objective: A closing discussion on BiomiX, the opportunities offered by multi-omics integration, and its current limitations, followed by a Q&A session.